From Predictive Agreement to Geometric Agreement: Stability of Fisher Connections and Semantic Transport in Neural Representations

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August 2026

Abstract

A probabilistic decoder equips its predictively identifiable hidden-state quotient with the pullback Fisher metric and Amari–Chentsov tensor. We establish a stability hierarchy for comparing this geometry across models. Exact equality of predictive images preserves every Amari α-connection, transport, curvature, and intrinsic predictive field. Uniform KL convergence alone does not imply even pointwise Fisher-metric convergence. General α-transport is stable under uniform interiority, nondegeneracy, and C^2 predictive agreement, but the resulting worst-case constants are poorly adapted to large vocabularies. For the Levi–Civita connection we remove both the vocabulary-size and minimum-token-probability dependence: C^2 closeness of the square-root predictive maps controls the connection and transport, and integrated KL plus a uniform Hilbert-valued Sobolev bound supplies this closeness at an explicit rate. A boundary Bernoulli family proves that the same square-root regularity cannot control any fixed nonzero-α connection without additional information. Score-moment identities identify the required raised third-order moment and yield probability-floor-free formulas at affine softmax heads. Finally, conditional Fisher variance is the exact obstruction to descending a semantic effect through a noninjective predictive map; it vanishes identically when that map is injective. Constant vector reuse at an affine head is exponential transport on the ambient intervention family, so superiority of any other connection is a held-out empirical hypothesis rather than a consequence of pretraining.

1 Introduction

Modern neural networks compute with vectors while producing probability distributions. For an autoregressive language model, a hidden state h entering the output head determines

$$F(h) = p(\cdot \mid h) \in \Delta_N^\circ, \tag{1}$$

where Δ_N° is the interior of the categorical simplex over a vocabulary of size N. The hidden coordinate remains a point of $\mathbb{R}^d$. Nevertheless, its predictively visible component parameterizes a statistical model and inherits the Fisher geometry of the decoder.

Exact equality of identifiable predictive images gives an isomorphism of their induced statistical structures. Approximate equality at the loss level is weaker: cross-entropy controls probability values on the data distribution, whereas Fisher metrics and affine connections depend on derivatives of the predictive map. Geometric equivalence also does not imply that a semantic operation is determined by the next-token distribution, because hidden states can contain decoder-null or semantically relevant fiber directions.

At a linear softmax head, constant hidden-vector reuse is exactly the affine operation selected by the flat exponential connection. Ordinary addition is therefore one connection-dependent rule rather than an alternative to geometry. The resulting question is which connection, if any, makes a specified semantic effect transferable across contexts and models.

The analysis proceeds through six results:

1. Exact predictive naturality: two identifiable parameterizations of the same predictive family preserve the Fisher metric, Amari-Chentsov tensor, every α-connection, parallel transport, and curvature up to reparameterization; every common smooth field on the shared predictive image pulls back equivariantly.

2. Separation of predictive and geometric convergence: a smooth, uniformly regular Bernoulli family converges uniformly in KL while its Fisher metric fails to converge at a fixed point.

3. Quantitative C^k stability: on compact uniformly interior and uniformly identifiable families, C^2 predictive closeness controls the defect between aligned α-connections and the commutator of their parallel transports, while C^3 closeness controls curvature.

4. A connection-selective boundary result: square-root regularity removes explicit vocabulary-size and minimum-probability factors from Levi–Civita bounds, but a Bernoulli family proves that it cannot control any fixed nonzero-α connection without an additional raised-third-moment assumption.

5. Loss-to-geometry implications under explicit regularity: integrated KL plus uniform Sobolev control of either probability coordinates or square-root coordinates yields transport convergence; the latter result is vocabulary independent for Levi–Civita transport.

6. Semantic conditional-variance decompositions: two orthogonal identities separately isolate failure to descend through a noninjective predictive map from connection mismatch and from cross-model conditional-mean-field mismatch.

These results define a concrete comparison protocol: choose a predictive quotient and a cross-model alignment, estimate infinitesimal semantic effects on a declared continuous chart, and compare exponential, Levi–Civita, mixture, and learned-α transport through commutation error and held-out behavioral transfer. The square-root sphere representation itself is classical; the contribution here is the quantitative implication–separation package and its decoder interpretation.

2 Related work

Information geometry equips regular statistical models with the Fisher metric, the Amari–Chentsov cubic tensor, and dual affine connections [2, 3, 6]. Invariance under sufficient statistics and the uniqueness of the Fisher and Amari–Chentsov tensors are classical [5, 8]. The square-root representation identifies the categorical Fisher metric with the metric induced from a Euclidean sphere. The distinction between affine flatness and Levi–Civita flatness is standard; global exponential or mixture coordinates do not force zero Riemannian curvature [7].

Pullback geometry in learned latent spaces is also established. Arvanitidis et al. [4] pull Fisher-Rao geometry through probabilistic decoders. Zhang [18] explicitly connects a language model's hidden-to-predictive map, likelihood training, and pullback geometry. For softmax representations, Park et al. [12] develop an information-geometric account of probing and steering, while Wang and

Zhao [15] pull the output Fisher metric through transformer layers and show large deviations from ambient Euclidean geometry. These results provide the decoder-side geometric setting used below.

Cross-model representation convergence has been studied from several directions. The Platonic Representation Hypothesis proposes convergence toward shared statistical structure [11]. Yu et al. [17] align distinct neural latent geometries with pullback metrics and relative geodesic representations. Hu et al. [10] model contextual transformations as vector fields and report shared displacement structure across language-model families. Path dependence and holonomy have also been proposed as representation diagnostics [14]. The stability analysis below asks when predictive agreement forces agreement of connection-dependent transport.

Continuity of Fisher information requires control of both distributions and their derivatives; Rezakhani et al. [13] prove such bounds in classical and quantum settings. Quantitative relaxations of sufficient statistics based on bi-Lipschitz Fisher metrics are studied by Yamaguchi and Nozawa [16]. Here a fixed comparison diffeomorphism aligns the predictive maps; their closeness is then propagated through every Amari α-connection to an explicit parallel-transport commutator bound. Sobolev regularity supplies the derivative control needed to derive this closeness from KL convergence.

3 Predictive statistical manifolds

3.1 The categorical simplex

Let

$$\Delta_N^\circ = \left\{ p \in \mathbb{R}^N : p_a > 0,\ \sum_{a=1}^N p_a = 1 \right\}. \tag{2}$$

For tangent vectors u, v, w satisfying $\sum_a u_a = \sum_a v_a = \sum_a w_a = 0$, define

$$g_p^\Delta(u, v) = \sum_{a=1}^N \frac{u_a v_a}{p_a}, \tag{3}$$

$$C_p^\Delta(u, v, w) = \sum_{a=1}^N \frac{u_a v_a w_a}{p_a^2}. \tag{4}$$

These are the Fisher metric and Amari-Chentsov tensor. A smooth predictive map

$$F : M \longrightarrow \Delta_N^\circ \tag{5}$$

induces

$$g_F = F^* g^\Delta, \qquad C_F = F^* C^\Delta. \tag{6}$$

When F is an immersion, g_F is a Riemannian metric. In local coordinates $x = (x^1, \dots, x^m)$, writing $p_a = F_a(x)$,

$$(g_F)_{ij} = \sum_a \frac{\partial_i p_a \, \partial_j p_a}{p_a}, \tag{7}$$

$$(C_F)_{ijk} = \sum_a \frac{\partial_i p_a \, \partial_j p_a \, \partial_k p_a}{p_a^2}. \tag{8}$$

Raise the final index of C_F using g_F and define K_F by

$$g_F(K_F(X, Y), Z) = C_F(X, Y, Z). \tag{9}$$

We use the convention

$$\nabla_F^{(\alpha)} = \nabla_F^{\mathrm{LC}} - \frac{\alpha}{2} K_F. \tag{10}$$

Thus $\alpha = 0$ is Levi-Civita, while $\alpha = 1$ and $\alpha = -1$ are the exponential and mixture connections on a regular exponential family.

3.2 Predictive quotients

Neural predictive maps are commonly nonidentifiable. Define

$$h \sim_F h' \iff F(h) = F(h'). \tag{11}$$

When the fibers are regular, the identifiable state space is the quotient $M_F = M/{\sim_F}$. All geometric statements below concern this quotient or a chosen horizontal subbundle on which the pullback metric is nondegenerate.

For a linear softmax decoder,

$$F(h) = \mathrm{softmax}(Wh + b), \tag{12}$$

the null space is

$$\mathcal{N} = \{u : Wu \in \mathrm{span}\{\mathbf{1}\}\}. \tag{13}$$

The quotient $\mathbb{R}^d/\mathcal{N}$ embeds smoothly into the simplex, and before quotienting

$$G(h) = W^\top(\operatorname{diag} p_h - p_h p_h^\top)W, \qquad \ker G(h) = \mathcal{N}. \tag{14}$$

This is the rigorous sense in which the predictively identifiable component of a hidden space is a statistical manifold. A finite set of naturally occurring hidden states remains a point cloud unless a smooth intervention chart or interpolation model is supplied.

3.3 Ordinary addition selects a connection

For the affine softmax family

$$p_h(a) = \exp\big(w_a^\top h + b_a - A(h)\big), \tag{15}$$

natural hidden coordinates are exponential-affine. Therefore constant coordinate vectors are $\nabla^{(1)}$-parallel. More explicitly,

$$p_{h_r + h_q - h_p}(a) = \frac{p_{h_r}(a) p_{h_q}(a)/p_{h_p}(a)}{\sum_b p_{h_r}(b) p_{h_q}(b)/p_{h_p}(b)}. \tag{16}$$

Thus hidden vector arithmetic in the ambient affine-head intervention family is exactly exponential transport, not the absence of geometry. Levi–Civita transport instead preserves Fisher lengths and angles, while mixture transport preserves the dual affine structure. Naturally occurring post-normalization states can occupy a lower-dimensional subset that is not closed under this arithmetic. Which connection best models a semantic operation on a declared intervention chart is an empirical question.

4 Exact predictive naturality

Exact predictive equivalence determines the corresponding metric, tensors, connections, and transports.

Theorem 4.1 (Exact predictive naturality)**.** *Let $F_A : M_A \to \Delta_N^\circ$ and $F_B : M_B \to \Delta_N^\circ$ be smooth embeddings onto the same embedded predictive image P. Define*

$$\Phi = F_B^{-1} \circ F_A : M_A \longrightarrow M_B. \tag{17}$$

Then

$$\Phi^* g_B = g_A, \qquad \Phi^* C_B = C_A. \tag{18}$$

For every $\alpha \in \mathbb{R}$,

$$\mathrm{d}\Phi(\nabla^{(\alpha)}_{A}{}_X Y) = \nabla^{(\alpha)}_{B}{}_{\mathrm{d}\Phi X}(\mathrm{d}\Phi Y). \tag{19}$$

For every piecewise C^1 path γ,

$$\mathrm{d}\Phi_{\gamma(1)} P^{A,\alpha}_{\gamma} = P^{B,\alpha}_{\Phi\circ\gamma} \mathrm{d}\Phi_{\gamma(0)}. \tag{20}$$

Curvature tensors intertwine, and holonomies of corresponding loops are conjugate.

Proof. Since $F_A = F_B \circ \Phi$, functoriality of pullback gives

$$\Phi^* g_B = \Phi^* F_B^* g^\Delta = (F_B \circ \Phi)^* g^\Delta = F_A^* g^\Delta = g_A, \tag{21}$$

and identically for C. A Riemannian isometry preserves the Levi-Civita connection. Preservation of g and C implies preservation of K, so (10) gives (19). Applying $\mathrm{d}\Phi$ to an A-parallel field and using uniqueness of the linear transport ODE gives (20). Curvature naturality follows from its definition as the commutator of covariant derivatives, and loop holonomies are consequently conjugate. □

Corollary 4.2 (Intrinsic semantic fields)**.** *Let $S_t : P \to P$ be a smooth semantic flow with infinitesimal field*

$$Z_P(p) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} S_t(p). \tag{22}$$

Define model fields X_i by $\mathrm{d}F_i(X_i) = Z_P \circ F_i$. Then

$$\mathrm{d}\Phi(X_A) = X_B \circ \Phi. \tag{23}$$

The condition is substantive: a context operation need not descend to P. If two hidden states have the same prediction before an intervention but distinct predictive effects afterward, then no intrinsic field Z_P exists on the predictive quotient. Section 8 quantifies this obstruction.

5 Prediction convergence does not imply geometric convergence

Let $P(\cdot \mid c)$ be the data conditional and $q(\cdot \mid c)$ a model. Population cross-entropy decomposes as

$$\mathcal{L}(q) - H_P(Y \mid C) = \mathbb{E}_C D_{\mathrm{KL}}\big(P(\cdot \mid C) \| q(\cdot \mid C)\big). \tag{24}$$

The logarithmic score is strictly proper [9]. If models q_A, q_B have excess risks ϵ_A, ϵ_B, Pinsker's inequality and the triangle inequality yield

$$\mathbb{E}_C \| q_A(\cdot \mid C) - q_B(\cdot \mid C)\|_1 \le \sqrt{2\epsilon_A} + \sqrt{2\epsilon_B}. \tag{25}$$

This makes output agreement natural under realizability and global population-risk optimization. It provides no derivative control.

Proposition 5.1 (Regular KL-to-geometry counterexample)**.** *There exist smooth, uniformly interior, regular Bernoulli predictive families p_n, p_* on $[-1,1]$ such that*

$$\sup_x D_{\mathrm{KL}}(p_*(x)\|p_n(x)) = O(n^{-2}), \tag{26}$$

but $g_{p_n}(0)$ does not converge to $g_{p_*}(0)$.

Proof. Let $p_*(x) = \mathrm{Ber}(r(x))$ and $p_n(x) = \mathrm{Ber}(r_n(x))$, where

$$r(x) = \frac{1}{3} + \frac{x}{20}, \qquad r_n(x) = r(x) + \frac{1}{100n}\sin(nx). \tag{27}$$

All probabilities remain in a fixed compact subset of $(0,1)$. Moreover,

$$r_n'(x) = \frac{1}{20} + \frac{1}{100}\cos(nx) \in [0.04, 0.06], \tag{28}$$

so every map is a regular immersion with a uniform rank bound. Uniform interiority and Taylor's theorem give $\sup_x D_{\mathrm{KL}}(p_*\|p_n) = O(n^{-2})$. The Bernoulli pullback metric is

$$g(x) = \frac{r'(x)^2}{r(x)(1-r(x))}\,\mathrm{d}x^2. \tag{29}$$

At $x = 0$, $r_n(0) = r(0) = 1/3$, but $r_n'(0) = 3/50$ and $r'(0) = 1/20$. Hence

$$g_n(0) = \frac{81}{5000}\,\mathrm{d}x^2 \neq \frac{9}{800}\,\mathrm{d}x^2 = g_*(0). \tag{30}$$

□

This example rules out an unconditional theorem of the form "cross-entropy convergence implies Fisher convergence." The oscillations have bounded first derivative but unbounded higher regularity, anticipating the Sobolev condition in Section 7.

There is a second nonidentifiability. For any hidden diffeomorphism ψ,

$$H' = \psi \circ H, \qquad F' = F \circ \psi^{-1} \tag{31}$$

produces exactly the same predictions while changing ambient Euclidean coordinates. Pullback Fisher geometry handles this gauge covariantly, but loss alone cannot identify a privileged hidden Euclidean metric.

6 Stability of Fisher geometry and transport

We now quantify how derivative-level predictive agreement propagates to geometry. We state the result on a common Euclidean chart. For two latent manifolds related by a comparison diffeomorphism Φ, replace the second map by $\widetilde{p} = F_B \circ \Phi$.

Let $K \Subset U \subset \mathbb{R}^m$ and let $p, \widetilde{p} : U \to \Delta_N^\circ$. For an integer k, use the componentwise norm

$$\|p\|_{C^k(K)} = \max_{a, |\beta| \le k} \sup_{x \in K} |\partial^\beta p_a(x)|. \tag{32}$$

Assumption 6.1 (Uniform regularity)**.** For constants $\eta, \lambda, B_2 > 0$,

$$p_a(x), \widetilde{p}_a(x) \geq \eta, \tag{33}$$

$$\|p\|_{C^2(K)} + \|\widetilde{p}\|_{C^2(K)} \leq B_2, \tag{34}$$

$$\lambda_{\min}(g_p(x)), \lambda_{\min}(g_{\widetilde{p}}(x)) \geq \lambda \tag{35}$$

for every a and $x \in K$.

The probability floor excludes the singular simplex boundary. The spectral floor excludes decoder-null directions after quotienting and prevents inverse-metric instability.

Theorem 6.2 (C^2 stability of α-transport)**.** *Under Assumption 6.1, fix $A > 0$. There is a constant*

$$C_\Gamma = C_\Gamma(m, N, A, \eta^{-1}, \lambda^{-1}, B_2) \tag{36}$$

such that, uniformly for $|\alpha| \leq A$,

$$\begin{aligned} &\|g_p - g_{\widetilde{p}}\|_{C^1(K)} + \|C_p - C_{\widetilde{p}}\|_{C^0(K)} \\ &\quad + \|\Gamma^{(\alpha)}_p - \Gamma^{(\alpha)}_{\widetilde{p}}\|_{C^0(K)} \leq C_\Gamma \|p - \widetilde{p}\|_{C^2(K)}. \end{aligned} \tag{37}$$

Let $\gamma : [0,1] \to K$ be piecewise C^1, with Euclidean length L_γ. If

$$M_\Gamma = \max\{\|\Gamma^{(\alpha)}_p\|_{C^0}, \|\Gamma^{(\alpha)}_{\widetilde{p}}\|_{C^0}\}, \tag{38}$$

where the norm is the induced bilinear operator norm, then

$$\|P^{p,\alpha}_\gamma - P^{\widetilde{p},\alpha}_\gamma\|_{\mathrm{op}} \leq L_\gamma e^{M_\Gamma L_\gamma} C_\Gamma \|p - \widetilde{p}\|_{C^2(K)}. \tag{39}$$

Proof. Equations (7) and (8) are finite sums of rational functions of p and its first derivatives. Their denominators are controlled by η. Differentiating (7) once uses at most second derivatives, giving the first two terms of (37) by the mean-value theorem on the compact regularity class.

For completeness, the lower-index α-coefficients have the direct probability-coordinate expression

$$\Gamma^{(\alpha)}_{ij,k} = \sum_a \left[\frac{\partial_{ij} p_a \, \partial_k p_a}{p_a} - \frac{1+\alpha}{2} \frac{\partial_i p_a \, \partial_j p_a \, \partial_k p_a}{p_a^2} \right]. \tag{40}$$

Thus their difference is $O(\|p - \widetilde{p}\|_{C^2})$. Raising the final index requires inverse metrics. The identity

$$g_p^{-1} - g_{\widetilde{p}}^{-1} = g_p^{-1}(g_{\widetilde{p}} - g_p) g_{\widetilde{p}}^{-1} \tag{41}$$

and the spectral floor λ give inverse-metric stability, proving (37).

Along γ, transport solves the linear ODE

$$\dot{V}(t) = -\Gamma^{(\alpha)}(\gamma(t))[\dot{\gamma}(t)]V(t). \tag{42}$$

Duhamel's formula bounds the difference of the two fundamental solutions by the integral of the coefficient defect multiplied by their growth factors. Gronwall's inequality yields (39). ☐

Corollary 6.3 (Cross-model transport commutator)**.** *Let $F_A : M_A \to \Delta_N^\circ$, $F_B : M_B \to \Delta_N^\circ$, and let $\Phi : M_A \to M_B$ be a C^2 local diffeomorphism. Suppose F_A and $F_B \circ \Phi$ satisfy Assumption 6.1 and*

$$\|F_A - F_B \circ \Phi\|_{C^2(K)} \le \varepsilon. \tag{43}$$

Then

$$\begin{aligned} &\|\mathrm{d}\Phi_{\gamma(1)} P_\gamma^{A,\alpha} - P_{\Phi\circ\gamma}^{B,\alpha} \mathrm{d}\Phi_{\gamma(0)}\|_{\mathrm{op}} \\ &\qquad \le L_\gamma e^{M_\Gamma L_\gamma} C_\Gamma \varepsilon, \end{aligned} \tag{44}$$

after using the aligned tangent norms. At $\varepsilon = 0$, this reduces to Theorem 4.1.

Proof. The connection induced by $F_B \circ \Phi$ is the pullback of the connection induced by F_B. Its transport is conjugated by $\mathrm{d}\Phi$. Apply Theorem 6.2 on M_A. □

Theorem 6.4 (C^3 curvature stability)**.** *Assume the hypotheses of Theorem 6.2, replace the C^2 bound by*

$$\|p\|_{C^3(K)} + \|\widetilde{p}\|_{C^3(K)} \le B_3, \tag{45}$$

and retain the probability and spectral floors. Then

$$\|R_p^{(\alpha)} - R_{\widetilde{p}}^{(\alpha)}\|_{C^0(K)} \le C_R \|p - \widetilde{p}\|_{C^3(K)}, \tag{46}$$

where C_R depends on $m, N, A, \eta^{-1}, \lambda^{-1}, B_3$. Levi-Civita sectional curvatures also converge on corresponding two-planes whose Gram determinants are uniformly bounded below.

Proof. Differentiating the argument for (37) gives C^1 control of the Christoffel symbols. Insert it into

$$R^\ell{}_{ijk} = \partial_i \Gamma^\ell{}_{jk} - \partial_j \Gamma^\ell{}_{ik} + \Gamma^\ell{}_{ir}\Gamma^r{}_{jk} - \Gamma^\ell{}_{jr}\Gamma^r{}_{ik}. \tag{47}$$

Sectional curvature is a quotient. The same estimates control its metric numerator, while the uniform Gram bound controls its denominator. □

6.1 Boundary-robust Levi–Civita stability

The preceding estimates are uniform over a probability-coordinate regularity class and are intentionally general in α. For Levi–Civita geometry, the square-root representation gives a sharper result. Define

$$\psi_p(x) = 2\big(\sqrt{p_1(x)}, \dots, \sqrt{p_N(x)}\big) \in \mathbb{R}^N. \tag{48}$$

Its image lies on the sphere of radius two. Equip target-valued derivatives with their Euclidean operator norms, and put

$$\|D^j\psi\|_{C^0(K),\mathrm{op}} = \sup_{\substack{x\in K\\ |u_1|,\dots,|u_j|\le 1}} \|D^j\psi_x[u_1,\dots,u_j]\|_{\ell_2}, \tag{49}$$

and put

$$\delta_j = \sup_{x\in K} \|D^j\psi_p(x) - D^j\psi_{\widetilde{p}}(x)\|_{\mathrm{op}}, \qquad j = 1, 2. \tag{50}$$

Theorem 6.5 (Vocabulary-independent Levi–Civita stability). *Suppose $\psi_p, \psi_{\widetilde{p}} : U \to \mathbb{R}^N$ are C^2 immersions and, on K,*

$$\begin{aligned} \|D\psi_p\|_{\mathrm{op}}, \|D\psi_{\widetilde{p}}\|_{\mathrm{op}} &\le B_1, \\ \|D^2\psi_p\|_{\mathrm{op}}, \|D^2\psi_{\widetilde{p}}\|_{\mathrm{op}} &\le B_2, \\ g_p, g_{\widetilde{p}} &\succeq \lambda I. \end{aligned} \tag{51}$$

Then

$$\|g_p - g_{\widetilde{p}}\|_{C^0(K)} \le 2B_1\delta_1, \tag{52}$$

$$\|Dg_p - Dg_{\widetilde{p}}\|_{C^0(K)} \le 2(B_1\delta_2 + B_2\delta_1), \tag{53}$$

$$\|\Gamma_p^{\mathrm{LC}} - \Gamma_{\widetilde{p}}^{\mathrm{LC}}\|_{C^0(K)} \le D_{\mathrm{LC}}, \tag{54}$$

where

$$D_{\mathrm{LC}} = \lambda^{-1}(B_1\delta_2 + B_2\delta_1) + 2B_1^2B_2\lambda^{-2}\delta_1. \tag{55}$$

For a path γ of Euclidean length L_γ,

$$\|P_\gamma^{p,\mathrm{LC}} - P_\gamma^{\widetilde{p},\mathrm{LC}}\|_{\mathrm{op}} \le L_\gamma D_{\mathrm{LC}} \min\left\{\exp\left(\frac{B_1B_2}{\lambda}L_\gamma\right), \frac{B_1^2}{\lambda}\right\}. \tag{56}$$

Every displayed constant is independent of N and of $\min_a p_a, \min_a \widetilde{p}_a$. The second transport branch uses metric compatibility and prevents an unavoidable exponential-in-length interpretation.

Proof. Differentiating (48) gives the exact identities

$$(g_p)_{ij} = \langle \partial_i\psi_p, \partial_j\psi_p\rangle, \qquad (\Gamma_p^{\mathrm{LC}})_{ij,k} = \langle \partial_{ij}\psi_p, \partial_k\psi_p\rangle. \tag{57}$$

The first is (7); the second follows either from the Koszul formula or by differentiating the first and cancelling the symmetric terms. Subtracting the two Gram matrices proves (52). Differentiating them once and applying the product rule proves (53). The lower-index coefficients in (57) differ by at most $B_1\delta_2 + B_2\delta_1$. Moreover,

$$\|g_p^{-1} - g_{\widetilde{p}}^{-1}\|_{\mathrm{op}} \le 2B_1\lambda^{-2}\delta_1, \tag{58}$$

and each lower-index coefficient has norm at most B_1B_2. Raising the final index yields (54)–(55). The same identities give $\|\Gamma^{\mathrm{LC}}\|_{\mathrm{op}} \le B_1B_2/\lambda$, so Duhamel's formula and Gronwall's inequality give the exponential branch of (56).

For the second branch, note that

$$\lambda I \preceq g_p, g_{\widetilde{p}} \preceq B_1^2 I. \tag{59}$$

Levi–Civita transport is an isometry for its own metric; hence every subpath propagator, measured in the background Euclidean norm, has operator norm at most $B_1/\sqrt{\lambda}$. The exact variation-of-connection identity contains one propagator from each geometry and one copy of their connection defect. Taking norms and integrating $|\dot\gamma|$ gives

$$\|P_\gamma^{p,\mathrm{LC}} - P_\gamma^{\widetilde{p},\mathrm{LC}}\|_{\mathrm{op}} \le L_\gamma \frac{B_1^2}{\lambda} D_{\mathrm{LC}}. \tag{60}$$

Taking the better of the two valid estimates proves (56). Appendix A gives the full Duhamel identity and its finite-scale interpretation. □

Corollary 6.6 (Boundary-robust Levi–Civita curvature)**.** *Under the hypotheses of Theorem 6.5, suppose additionally that the two square-root immersions are C^3 so that their classical curvature tensors are defined. Then the covariant Levi–Civita curvature tensors satisfy*

$$\| \mathrm{Rm}_p - \mathrm{Rm}_{\widetilde{p}} \|_{C^0(K)} \le C(B_1, B_2, \lambda^{-1})(\delta_1 + \delta_2). \tag{61}$$

Consequently, sectional curvatures converge on corresponding planes whose Fisher Gram determinants are uniformly bounded below. The constant is again independent of vocabulary size and minimum token probability.

Proof. Regard each square-root map as a Euclidean immersion. Its second fundamental form is

$$\mathrm{II}_p(u, v) = (I - \Pi_p) D^2 \psi_p(u, v), \qquad \Pi_p = D\psi_p \, g_p^{-1} (D\psi_p)^*. \tag{62}$$

The inverse-metric estimate used above controls $\Pi_p - \Pi_{\widetilde{p}}$ by $C(B_1, \lambda^{-1})\delta_1$, and hence controls the second fundamental forms by $C(B_1, B_2, \lambda^{-1})(\delta_1 + \delta_2)$. The Euclidean Gauss equation expresses Rm as a quadratic contraction of II, proving the tensor bound. The sectional-curvature statement follows by controlling its metric denominator. $\square$

6.2 Score moments and the affine-head specialization

The absence of explicit boundary factors above is not an algebraic accident. Let $Y \sim p(x)$ and define the score and its derivative by

$$S_i(Y, x) = \partial_i \log p_Y(x), \qquad H_{ij}(Y, x) = \partial_i S_j(Y, x). \tag{63}$$

Proposition 6.7 (Score-moment form of the α-connection)**.** *For every smooth categorical family,*

$$(g_p)_{ij} = \mathbb{E}_p[S_i S_j], \tag{64}$$

$$(C_p)_{ijk} = \mathbb{E}_p[S_i S_j S_k], \tag{65}$$

$$\Gamma^{(\alpha)}_{ij,k} = \mathbb{E}_p[H_{ij} S_k] + \frac{1-\alpha}{2} \mathbb{E}_p[S_i S_j S_k]. \tag{66}$$

For an affine softmax head in a common natural-coordinate chart, let G, C and $\widetilde{G}, \widetilde{C}$ denote the corresponding second and third centered moments of the decoder rows. Then

$$\Gamma^{(\alpha)} = \frac{1-\alpha}{2} G^{-1} C. \tag{67}$$

If $G, \widetilde{G} \succeq \lambda I$, then

$$\begin{aligned} \|\Gamma^{(\alpha)} - \widetilde{\Gamma}^{(\alpha)}\|_{\mathrm{op}} \le \frac{|1-\alpha|}{2} \Big(& \lambda^{-1} \|C - \widetilde{C}\|_{\mathrm{op}} \\ & + \lambda^{-2} \|G - \widetilde{G}\|_{\mathrm{op}} \|\widetilde{C}\|_{\mathrm{op}} \Big). \end{aligned} \tag{68}$$

This bound has no explicit dependence on vocabulary size or minimum token probability.

Proof. Since $\partial_i p_a = p_a S_i(a)$ and

$$\partial_{ij} p_a = p_a \big(H_{ij}(a) + S_i(a) S_j(a) \big), \tag{69}$$

substitution into (7), (8), and (40) proves (64)–(66). At an affine head, $S(Y) = w_Y - \mathbb{E}_p w_Y$ and $H_{ij}(Y) = -G_{ij}$ is independent of Y. Because $\mathbb{E}_p S = 0$, the first term of (66) vanishes, giving (67). Finally use

$$G^{-1} C - \widetilde{G}^{-1} \widetilde{C} = G^{-1}(C - \widetilde{C}) + (G^{-1} - \widetilde{G}^{-1}) \widetilde{C} \tag{70}$$

and $\|G^{-1} - \widetilde{G}^{-1}\|_{\mathrm{op}} \le \lambda^{-2} \|G - \widetilde{G}\|_{\mathrm{op}}$. $\square$

Corollary 6.8 (General α stability from cubic-moment control)**.** *In the setting of Theorem 6.5, suppose additionally that*

$$\|C_p\|_{C^0(K),\mathrm{op}}, \|C_{\widetilde{p}}\|_{C^0(K),\mathrm{op}} \le M_3, \qquad \|C_p - C_{\widetilde{p}}\|_{C^0(K),\mathrm{op}} \le \delta_C. \tag{71}$$

Then

$$\|\Gamma_p^{(\alpha)} - \Gamma_{\widetilde{p}}^{(\alpha)}\|_{\mathrm{op}} \le D_{\mathrm{LC}} + \frac{|\alpha|}{2}\left(\lambda^{-1}\delta_C + 2B_1M_3\lambda^{-2}\delta_1\right). \tag{72}$$

Thus square-root C^2 control alone determines Levi–Civita stability; a general nonzero-α connection additionally requires stability of the raised third score moment. A tokenwise probability floor is one sufficient route, not a necessary one. Affine softmax structure supplies a floor-free route directly through Proposition 6.7.

Proof. Use $\Gamma^{(\alpha)} = \Gamma^{\mathrm{LC}} - (\alpha/2)g^{-1}C$, Theorem 6.5, and the inverse-metric estimate in its proof. □

Proposition 6.9 (Square-root control does not control nonzero α)**.** *There is a one-dimensional categorical family whose square-root map has uniformly bounded derivatives of every order and whose Fisher metric is identically one, while every fixed nonzero-α connection coefficient diverges toward the simplex boundary.*

Proof. For $0 < \theta < \pi$, take the Bernoulli family

$$p_\theta = \left(\sin^2\frac{\theta}{2}, \cos^2\frac{\theta}{2}\right). \tag{73}$$

Its square-root representation is

$$\psi_\theta = 2\left(\sin\frac{\theta}{2}, \cos\frac{\theta}{2}\right). \tag{74}$$

All derivatives are bounded as $\theta \downarrow 0$, while

$$g_{\theta\theta} = 1, \qquad \Gamma^{\mathrm{LC}}_{\theta\theta,\theta} = \langle \psi''_\theta, \psi'_\theta\rangle = 0. \tag{75}$$

The two score values are $\cot(\theta/2)$ and $-\tan(\theta/2)$, so

$$C_{\theta\theta\theta} = 2\cot\theta. \tag{76}$$

Consequently,

$$\Gamma^{(\alpha)}_{\theta\theta,\theta} = -\alpha\cot\theta, \tag{77}$$

which diverges for every $\alpha \neq 0$. Thus square-root C^2 control and a perfect Fisher spectral floor are sufficient for Levi–Civita stability but not for a general α-connection. At an affine softmax head in natural coordinates, Proposition 6.7 supplies a separate moment-based route and $\Gamma^{(1)} = 0$ identically. □

Remark 6.10 (Why the assumptions matter)*.* The probability-coordinate constants in Theorems 6.2 and 6.4 diverge near $p_a = 0$ because their uniform class controls derivatives of p_a independently of p_a. This is not always an intrinsic blow-up of the pulled-back geometry: Theorem 6.5 captures cancellations through square-root derivatives, while Proposition 6.7 captures them through probability-weighted score moments. Rank loss remains intrinsic because raising indices requires g^{-1}. For nonzero α, the raised third score moment supplies information not determined by the Fisher metric alone and must be controlled explicitly; uniform interiority is sufficient but not logically necessary. Vocabulary independence means that no explicit N occurs in the displayed constants; the derivative bounds and spectral floor must still be uniform and may deteriorate with model scale. The estimates are conditional finite-scale bounds, not numerical certificates until those quantities are measured.

7 From cross-entropy to transport under Sobolev regularity

Theorem 6.2 assumes derivative-level agreement. We now give sufficient conditions under which population KL convergence implies it.

Let Ω be a compact m-dimensional smooth manifold or a bounded Sobolev-extension domain. Fix a smooth background Riemannian metric on the former and the Euclidean metric on the latter; all volume measures, Sobolev norms, operator norms, and path lengths in this section refer to that background structure. Let μ have density ρ with respect to volume, satisfying $\rho \geq \rho_0 > 0$. Define

$$\mathcal{K}(p, \widetilde{p}) = \int_\Omega D_{\mathrm{KL}}(p(x) \| \widetilde{p}(x)) \, \mathrm{d}\mu(x). \tag{78}$$

Theorem 7.1 (Risk-to-transport convergence). *Let $p, \widetilde{p} : \Omega \to \Delta_N^\circ$ satisfy the probability and spectral floors of Assumption 6.1. Suppose*

$$p, \widetilde{p} \in H^s(\Omega; \mathbb{R}^N), \qquad s > 2 + \frac{m}{2}, \tag{79}$$

and

$$\|p\|_{H^s} + \|\widetilde{p}\|_{H^s} \leq B. \tag{80}$$

Fix $A, L_0 > 0$. For every r with

$$2 + \frac{m}{2} < r < s, \tag{81}$$

there is a constant $C = C(\Omega, m, s, r, N, A, L_0, \rho_0^{-1}, \eta^{-1}, \lambda^{-1}, B)$ such that, uniformly for $|\alpha| \leq A$ and every path γ of background length $L_\gamma \leq L_0$,

$$\|P_\gamma^{p,\alpha} - P_\gamma^{\widetilde{p},\alpha}\|_{\mathrm{op}} \leq C B^{r/s} \mathcal{K}(p, \widetilde{p})^{\frac{s-r}{2s}}. \tag{82}$$

Equivalently, the transport error is $O(\mathcal{K}^\beta)$ for every

$$0 < \beta < \frac{s - 2 - m/2}{2s}. \tag{83}$$

Proof. Pinsker's inequality and $\rho \geq \rho_0$ give

$$\begin{aligned} \|p - \widetilde{p}\|_{L^2(\mathrm{dvol})}^2 &\leq \rho_0^{-1} \int_\Omega \|p - \widetilde{p}\|_1^2 \, \mathrm{d}\mu \\ &\leq 2\rho_0^{-1} \mathcal{K}(p, \widetilde{p}). \end{aligned} \tag{84}$$

Sobolev interpolation [1] yields

$$\|p - \widetilde{p}\|_{H^r} \leq C \|p - \widetilde{p}\|_{L^2}^{1-r/s} \|p - \widetilde{p}\|_{H^s}^{r/s}, \tag{85}$$

and therefore

$$\|p - \widetilde{p}\|_{H^r} \leq C B^{r/s} \mathcal{K}(p, \widetilde{p})^{(s-r)/(2s)}. \tag{86}$$

Since $r > 2 + m/2$, Sobolev embedding gives $H^r \hookrightarrow C^2$. Apply Theorem 6.2. ☐

Theorem 7.2 (Boundary-robust risk-to-Levi–Civita transport). *Let $p, \widetilde{p} : \Omega \to \Delta_N^\circ$, write $\psi_p = 2\sqrt{p}$ and $\psi_{\widetilde{p}} = 2\sqrt{\widetilde{p}}$ componentwise, and assume*

$$\psi_p, \psi_{\widetilde{p}} \in H^s(\Omega; \ell_2^N), \qquad s > 2 + \frac{m}{2}, \tag{87}$$

with

$$\|\psi_p\|_{H^s} + \|\psi_{\widetilde{p}}\|_{H^s} \le B, \qquad g_p, g_{\widetilde{p}} \succeq \lambda I. \tag{88}$$

Fix $L_0 > 0$. For every r satisfying $2 + m/2 < r < s$ and every path of background length at most L_0,

$$\|P_\gamma^{p,\mathrm{LC}} - P_\gamma^{\widetilde{p},\mathrm{LC}}\|_{\mathrm{op}} \le C\,\mathcal{K}(p,\widetilde{p})^{\frac{s-r}{2s}}, \tag{89}$$

where C depends on $\Omega, s, r, \rho_0^{-1}, B, \lambda^{-1}, L_0$ but not on N or a minimum token probability. Equivalently, the transport error is $O(\mathcal{K}^\beta)$ for every

$$0 < \beta < \frac{s - 2 - m/2}{2s}. \tag{90}$$

Proof. For categorical distributions the scalar inequality

$$\sum_a (\sqrt{p_a} - \sqrt{\widetilde{p}_a})^2 \le D_{\mathrm{KL}}(p\|\widetilde{p}) \tag{91}$$

gives

$$\|\psi_p - \psi_{\widetilde{p}}\|^2_{L^2(\mathrm{dvol};\ell_2)} \le 4\rho_0^{-1}\mathcal{K}(p,\widetilde{p}). \tag{92}$$

Hilbert-valued Sobolev interpolation and embedding give

$$\begin{aligned} \|\psi_p - \psi_{\widetilde{p}}\|_{C^2(\ell_2)} &\le C\|\psi_p - \psi_{\widetilde{p}}\|_{H^r(\ell_2)} \\ &\le C B^{r/s}\mathcal{K}(p,\widetilde{p})^{(s-r)/(2s)}. \end{aligned} \tag{93}$$

The same embedding bounds the first two derivatives of each square-root map in terms of B. Apply Theorem 6.5. □

Corollary 7.3 (Two-model geometric convergence)**.** *Suppose two model sequences $p_{A,n}, p_{B,n}$ on a shared intervention chart converge in population cross-entropy to the same conditional map p_*, and all three maps satisfy uniform versions of the hypotheses of Theorem 7.1. Then, for every admissible β,*

$$\|P_\gamma^{A,n,\alpha} - P_\gamma^{B,n,\alpha}\|_{\mathrm{op}} \le C(\epsilon_{A,n}^\beta + \epsilon_{B,n}^\beta), \tag{94}$$

where $\epsilon_{i,n} = \mathbb{E}D_{\mathrm{KL}}(p_*\|p_{i,n})$. With a cross-model alignment Φ_n, the same statement holds for the transport commutator in (44) after replacing $p_{B,n}$ by $p_{B,n} \circ \Phi_n$.

Alternatively, if the square-root maps satisfy uniform versions of Theorem 7.2, their Levi–Civita transports obey the same conclusion with constants independent of vocabulary size and minimum token probability.

If $s > 3 + m/2$, the same argument combined with Theorem 6.4 gives curvature convergence for every

$$0 < \beta < \frac{s - 3 - m/2}{2s}. \tag{95}$$

For Levi–Civita curvature, Corollary 6.6 instead uses C^2 convergence of the square-root immersions. Thus Theorem 7.2 also gives vocabulary-independent LC curvature convergence at the transport exponent, provided additionally that the square-root immersions are C^3 so their classical curvature tensors are defined and that the compared plane Gram determinants stay uniformly positive.

The uniform Sobolev hypothesis excludes the high-frequency behavior in Proposition 5.1. For language models it should be tested on low-dimensional continuous intervention charts, such as soft-prompt paths or controlled activation subspaces, rather than assumed on the discrete set of tokenized contexts.

8 The semantic-sufficiency obstruction

Even perfect agreement of predictive geometry does not guarantee that a semantic operation is a field on that geometry. We now separate this representational obstruction from connection mismatch.

Let $F : \mathcal{H} \to P$ be a regular predictive quotient, let H be a random hidden state with law μ, and let $X_T(H) \in T_H\mathcal{H}$ describe a semantic operation T. Its observable infinitesimal predictive effect is

$$Z_T(H) = \mathrm{d}F_H X_T(H) \in T_{F(H)}P. \tag{96}$$

Assume square integrability in the Fisher metric. Since P is embedded in the simplex, tangent-valued conditional expectations may be taken fiberwise in the ambient zero-sum space; conditioning on $F(H) = p$ fixes both T_pP and g_p.

Theorem 8.1 (Semantic conditional-variance obstruction)**.** *For a measurable, square-integrable field $Y : P \to TP$, define*

$$\mathcal{E}_T(Y) = \mathbb{E}\|Z_T(H) - Y(F(H))\|^2_{g_{F(H)}}. \tag{97}$$

Let

$$\overline{Z}_T(p) = \mathbb{E}[Z_T(H) \mid F(H) = p]. \tag{98}$$

Then

$$\begin{aligned}\mathcal{E}_T(Y) = \mathbb{E}\|Z_T - \overline{Z}_T(F(H))\|^2_g \\ + \mathbb{E}\|\overline{Z}_T(F(H)) - Y(F(H))\|^2_g.\end{aligned} \tag{99}$$

Consequently,

$$\inf_Y \mathcal{E}_T(Y) = \mathbb{E}\,\mathrm{Var}_g(Z_T \mid F). \tag{100}$$

The semantic effect descends almost surely to a predictive field if and only if this conditional variance is zero. Smooth descent additionally requires a smooth version of $\overline{Z}_T$.

Proof. Square-integrable tangent-valued random variables form a Hilbert direct integral with inner product

$$\langle U, V\rangle = \mathbb{E}[g_{F(H)}(U, V)]. \tag{101}$$

The fields measurable with respect to $\sigma(F)$ form a closed subspace. Conditional expectation is the orthogonal projection onto that subspace. Equivalently, decompose

$$\begin{aligned}Z_T - Y(F) = \big(Z_T - \overline{Z}_T(F)\big) \\ + \big(\overline{Z}_T(F) - Y(F)\big).\end{aligned} \tag{102}$$

The two terms are orthogonal after conditioning on F, which proves (99) and (100). □

Corollary 8.2 (Injective maps have no quotient obstruction)**.** *If F is injective on the support of H and its inverse on $F(\operatorname{supp} H)$ is measurable, then*

$$\mathrm{Var}_g(Z_T \mid F) = 0 \qquad \textit{almost surely.} \tag{103}$$

For the affine softmax decoder, this applies whenever

$$\mathcal{N} = \{u : Wu \in \operatorname{span}\{\mathbf{1}\}\} = \{0\}, \tag{104}$$

equivalently whenever the pullback Fisher matrix (14) is positive definite.

Proof. Under the stated injectivity, $H = F^{-1}(F(H))$ almost surely, so every measurable function of H, including $Z_T(H)$, is measurable with respect to $\sigma(F(H))$. Its conditional variance is therefore zero. The affine-softmax equivalence follows from

$$F(h) = F(h') \quad \iff \quad W(h - h') \in \operatorname{span}\{\mathbf{1}\}. \tag{105}$$

□

Corollary 8.2 is a scope condition, not a defect of Theorem 8.1. The obstruction is informative at a pre-quotient representation with nontrivial predictive fibers or at an earlier layer whose remaining network is noninjective. It is identically inert at a minimal full-rank affine head. If a smooth coarsening $S : P \to \bar{P}$ is declared, the theorem applies anew to $\bar{F} = S \circ F$, the effect $\mathrm{d}\bar{F}\, X_T$, and the metric and tangent bundle of the coarsened image. Merely conditioning the original P-tangent field on $S(F)$ can mix distinct tangent spaces and is not covered by the theorem. Approximate conditioning or finite bins likewise define a different, resolution-dependent diagnostic.

Corollary 8.3 (Connection-restricted decomposition)**.** *Let $\mathcal{C}_\alpha$ be any closed class of predictive fields selected by an α-connection, for example fields satisfying $\nabla^{(\alpha)} Y = 0$ on a specified simply connected chart or fields with a bounded covariant-derivative residual. Then*

$$\begin{aligned} \inf_{Y \in \mathcal{C}_\alpha} \mathcal{E}_T(Y) = {} & \underbrace{\mathbb{E}\, \mathrm{Var}_g(Z_T \mid F)}_{\textit{predictive insufficiency}} \\ & + \underbrace{\inf_{Y \in \mathcal{C}_\alpha} \mathbb{E} \| \overline{Z}_T - Y \|_g^2}_{\textit{connection mismatch}} . \end{aligned} \tag{106}$$

Corollary 8.4 (Cross-model semantic transfer)**.** *Let P_A, P_B be predictive quotients related by a C^1 diffeomorphism $\Phi : P_A \to P_B$ on the relevant supports. Write $Q_i = F_i(H_i)$ for the random predictive state of model i. Define*

$$Y_{A \to B}(p_B) = \mathrm{d}\Phi_{\Phi^{-1}(p_B)} \overline{Z}_A(\Phi^{-1}(p_B)). \tag{107}$$

Then

$$\begin{aligned} \mathbb{E} \| Z_B - Y_{A \to B}(Q_B) \|_{g_B}^2 = {} & \underbrace{\mathbb{E} \| Z_B - \overline{Z}_B(Q_B) \|_{g_B}^2}_{\textit{model-B quotient obstruction}} \\ & + \underbrace{\mathbb{E} \| \overline{Z}_B - \mathrm{d}\Phi\, \overline{Z}_A \|_{g_B}^2}_{\textit{cross-model field mismatch}} . \end{aligned} \tag{108}$$

Identity (108) is measured in g_B and requires only that the transferred field be well defined and measurable. If Φ is additionally a Fisher isometry, the mismatch has an intrinsic metric-preserving interpretation. Otherwise it is relative to the proposed alignment and its distortion must be reported separately.

No choice of α can reduce the first term of (106). On a genuinely noninjective predictive map, a strictly positive population conditional Fisher variance therefore proves that no predictive-state-only field, and hence no choice of α-transport on that quotient, can represent the specified infinitesimal effect exactly. At an injective final head the term vanishes tautologically and is not an empirical diagnostic.

9 Operational tests and training implications

The theorems suggest a hierarchy of increasingly strong cross-model agreement tests:

$$\begin{aligned}
&\text{Level } 0 &&: \text{output KL agreement,} \\
&\text{Level } 1 &&: \text{Fisher metric agreement,} \\
&\text{Level } 2 &&: \alpha\text{-connection and transport agreement,} \\
&\text{Level } 3 &&: \text{semantic-field equivariance,} \\
&\text{Level } 4 &&: \text{curvature and holonomy agreement.}
\end{aligned} \tag{109}$$

Agreement at one level does not imply the next without the regularity and sufficiency conditions identified above.

9.1 Transport commutation score

For an estimated alignment Φ and sampled paths and tangent vectors, define

$$\mathcal{D}_{\mathrm{PT}}^{(\alpha)}(A, B; \Phi) = \mathbb{E}_{\gamma,v} \frac{\|\mathrm{d}\Phi_{\gamma(1)} P_{\gamma}^{A,\alpha} v - P_{\Phi\gamma}^{B,\alpha} \mathrm{d}\Phi_{\gamma(0)} v\|}{\|v\|}. \tag{110}$$

Theorem 4.1 makes this score zero under exact predictive equivalence, while Corollary 6.3 bounds it under approximate equivalence. A practical study should test whether (110) predicts held-out steering transfer or task generalization beyond output KL and representation-only similarities such as CKA.

9.2 Selecting the connection

At a linear softmax head, $\alpha = 1$ is ordinary vector reuse, $\alpha = 0$ is Fisher-metric-compatible Levi-Civita transport, and $\alpha = -1$ is mixture transport. A predictive semantic field Y_T can be evaluated through

$$\mathcal{R}_T(\alpha) = \mathbb{E}\|\nabla^{(\alpha)} Y_T\|_g^2. \tag{111}$$

The minimizer is interpreted relative to the tested field and chart:

- a minimum at the exponential connection means that the tested field is closest to exponential-parallel; at the affine softmax head this is constant natural-coordinate vector reuse;
- a minimum at the Levi-Civita connection means that the tested field is closest to the metric-compatible transport;
- a minimum at the mixture or an intermediate α identifies the corresponding affine connection as the closest member of the tested family; intermediate α-connections need not be flat;
- on a verified noninjective predictive map, positive population conditional variance proves predictive insufficiency independently of the minimizing connection;
- on such a map, small estimated conditional variance together with large residuals for every tested α motivates an operation-specific connection or a nonparallel field model.

The construction of the tested field must be declared separately from the connection comparison. A finite hidden difference $h(Tc) - h(c)$ is the exponential logarithm of its endpoints at an affine head and therefore intentionally tests transfer of the conventional exponential-coordinate intervention. A connection-neutral infinitesimal test instead specifies a continuous intervention T_t and uses

$$X_T(h(c)) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} h(T_t c). \tag{112}$$

For finite endpoints, each candidate can be evaluated with its own $\mathrm{Log}^{(\alpha)}$–transport–$\mathrm{Exp}^{(\alpha)}$ construction. Any minimizing $\alpha \neq 1$ remains a descriptive best fit within the tested family; it does not by itself identify a mechanism implemented by optimization.

9.3 Training regimes

The risk-to-transport theorems indicate why derivative regularity matters. A causal training study can compare cross-entropy-only models with matched-loss models trained using

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{CE}} + \lambda \widehat{\mathrm{Var}}_g(\mathrm{d}FX_T \mid F) + \mu \|\nabla^{(\alpha)} \overline{Z}_T\|_g^2 + \nu \mathcal{R}_{\mathrm{smooth}}. \tag{113}$$

The first regularizer is included only for a declared noninjective predictive map and encourages the operation to descend through that map. Under coarsening, every occurrence of F, $\mathrm{d}FX_T$, and g in this term must be replaced by $\bar{F} = S \circ F$, $\mathrm{d}\bar{F}X_T$, and the coarsened-image metric. It must be omitted at an injective final head, where it is identically zero. The second selects connection-relative consistency. For Levi–Civita transport, the final term should control the square-root predictive map in the Sobolev norm used below; for general α, it must additionally control the required score moments. Applying either theorem requires verifying the corresponding uniform bound. The empirical protocol treats behavioral improvement as the primary outcome and the geometric regularizers as diagnostic quantities.

A no-GPU pilot can use frozen small models and one- or two-dimensional soft-prompt charts. Finite differences estimate F, ∂F, and $\partial^2 F$; the model's output head provides exact vocabulary probabilities. The main comparisons are output KL, metric defect, the transport score (110), connection residual, and held-out intervention transfer. Conditional variance is added only at a representation with verified nontrivial predictive fibers or after a complete coarsened map, effect, and image geometry are explicitly defined.

An inferential curvature study must also separate semantic structure from Fisher-spectrum alignment. Isotropic Fisher–Haar planes answer a literal isotropic null but are not spectrum matched. A task-specific analysis should preserve the semantic plane's leverage across preregistered Fisher spectral bands, using signs for singleton bands and basis-invariant Haar rotations inside repeated or clustered bands. Its plus-one control-tail rank is a calibrated randomization p-value only under an explicit conditional invariance null for that block-orthogonal group; direction and multiplicity must also be preregistered. First-order α fits must be extrapolated to zero edge length or replaced by integrated transport; their finite-edge parameter error is generically first order in Fisher length.

10 Limitations

Continuous charts. Tokenized contexts are discrete. Derivatives and connections require an explicit continuous chart, such as soft prompts, activation interventions, or a justified local manifold model.

Boundary and rank. Softmax probabilities are positive but can be extremely small. Uniform tokenwise floors are mathematically clean and practically strong, and the general-α probability-coordinate bound is not a useful finite-scale certificate when its constants are dominated by that floor. The square-root Levi–Civita theorem removes this explicit dependence, and affine heads admit score-moment formulas, but Fisher rank loss remains a genuine instability. Quotienting, coarse-graining, or a preregistered horizontal-subspace analysis is necessary near rank loss; an unreported ridge changes the geometry.

Architectural support. The ambient affine-head family permits arbitrary interventions in its chosen hidden coordinates. Natural post-LayerNorm states satisfy architectural constraints and need not fill that family or remain on their support after vector addition. Claims about natural-state geometry therefore require a chart tangent to the reachable support or a pullback through the normalization map; ambient intervention geometry and support-intrinsic geometry must not be conflated.

Different tokenizers. The exact theorem assumes a common outcome space. A tokenizer map generally acts as a Markov morphism. Fisher geometry is preserved only under a statistically sufficient or reversible identification; otherwise it contracts [5].

Population versus empirical loss. Equation (24) concerns population risk, realizability, and global optimization. Finite empirical loss introduces estimation and optimization errors not treated here.

Alignment is not identified automatically. The stability theorem evaluates a proposed Φ. Constructing a statistically identifiable alignment from finite observations, especially across different latent dimensions, is a separate problem. One may instead work on a shared low-dimensional intervention chart.

Connection dependence. Parallel transport compares tangent vectors at different points relative to a selected connection. At the final affine softmax head, ambient addition coincides with exponential-affine transport, while Levi-Civita and mixture transport preserve different structures. The theory therefore evaluates connection choice separately across operations, layers, and tasks.

11 Conclusion

Idealized pretraining makes convergence of conditional output distributions natural because excess cross-entropy is expected KL divergence. It does not by itself make convergence of Fisher metrics, affine connections, or curvature natural. Geometry depends on derivatives, and a regular oscillatory counterexample separates these notions.

Under explicit smoothness and identifiability assumptions, stronger conclusions become provable. Predictive maps that converge in C^2 have convergent Amari α-connections and parallel transports; C^3 convergence controls curvature. For Levi–Civita transport, square-root coordinates give vocabulary-independent bounds without a uniform probability floor, and integrated KL plus a uniform Hilbert-valued Sobolev bound supplies the missing derivative convergence. Nonzero-α connections require additional raised-third-score-moment control, for which a tokenwise probability floor is sufficient but not necessary. The semantic conditional-variance obstruction determines whether a noninjective predictive map is rich enough to support the operation; at an injective head it vanishes by construction.

If aligned predictive maps have vanishing population KL risk together with the uniform Sobolev, boundary, and rank bounds of Theorem 7.1, their α-transport commutators vanish. Semantic-field equivariance is a separate condition: it requires small within-model quotient variance and small cross-model conditional-mean-field mismatch. These quantities localize deviations from transfer to inadequate regularity, predictive insufficiency, field mismatch, or connection mismatch.

A Square-root boundary calculus and finite-scale auditing

A.1 What the square-root sphere does and does not flatten

The classical square-root map $\psi = 2\sqrt{p}$ sends the open categorical simplex to the positive orthant of the radius-two sphere and satisfies

$$\sum_a \psi_a^2 = 4, \qquad g = D\psi^* D\psi. \tag{114}$$

The ambient inner product is Euclidean, but the induced Fisher geometry is not globally Euclidean: the full simplex has Levi–Civita sectional curvature 1/4. Thus the sphere representation supplies a Gram factorization of the metric, not a proof of intrinsic flatness.

The metric and first-kind Levi–Civita coefficients,

$$g_{ij} = \langle \partial_i \psi, \partial_j \psi \rangle, \qquad \Gamma^{\mathrm{LC}}_{ij,k} = \langle \partial_{ij}\psi, \partial_k \psi \rangle, \tag{115}$$

are polynomial contractions of first and second immersion derivatives. Raised coefficients and tangent projections still contain g^{-1}, so the square-root representation removes explicit tokenwise denominators but does not remove Fisher rank conditioning.

For the cubic tensor, $p_a = \psi_a^2/4$ and $\partial_i p_a = (\psi_a/2)\partial_i \psi_a$ give the exact identity

$$C_{ijk} = 2 \sum_a \frac{\partial_i \psi_a \, \partial_j \psi_a \, \partial_k \psi_a}{\psi_a}. \tag{116}$$

Consequently, square-root C^2 bounds alone do not uniformly bound C near a vanishing component. Equation (116) does not say that C must diverge: its numerator can vanish sufficiently quickly, or structural score-moment identities can provide cancellation.

Proposition A.1 (Nonzero-α stability criterion)**.** *On a common chart, put $K_p = g_p^{-1} C_p$ and $K_{\widetilde{p}} = g_{\widetilde{p}}^{-1} C_{\widetilde{p}}$. For every fixed $\alpha \neq 0$,*

$$\|\Gamma^{(\alpha)}_p - \Gamma^{(\alpha)}_{\widetilde{p}}\| \le \|\Gamma^{\mathrm{LC}}_p - \Gamma^{\mathrm{LC}}_{\widetilde{p}}\| + \frac{|\alpha|}{2} \|K_p - K_{\widetilde{p}}\|, \tag{117}$$

$$\|K_p - K_{\widetilde{p}}\| \le \frac{2}{|\alpha|} \left(\|\Gamma^{(\alpha)}_p - \Gamma^{(\alpha)}_{\widetilde{p}}\| + \|\Gamma^{\mathrm{LC}}_p - \Gamma^{\mathrm{LC}}_{\widetilde{p}}\| \right). \tag{118}$$

Hence, once Levi–Civita stability is known, stability of one fixed nonzero-α connection is equivalent to stability of the raised third score moment. No tokenwise probability floor is logically necessary.

Proof. Both estimates are the triangle inequality applied in opposite directions to

$$\Gamma^{(\alpha)} = \Gamma^{\mathrm{LC}} - \frac{\alpha}{2} K. \tag{119}$$

□

At an affine softmax head, write

$$S_a(u) = (w_a - \mathbb{E}_p w_Y)^\top u. \tag{120}$$

Then the first two derivatives of the square-root map satisfy

$$D\psi_a[u] = \sqrt{p_a}\, S_a(u), \tag{121}$$

$$D^2\psi_a[u,v] = \sqrt{p_a}\left(\frac{1}{2}S_a(u)S_a(v) - g(u,v)\right). \tag{122}$$

Because $\mathbb{E}_p[S(u)S(v)] = g(u,v)$, expansion of the squared norm gives

$$\|D^2\psi[u,v]\|_{\ell_2}^2 = \frac{1}{4}\mathbb{E}_p[S_Y(u)^2 S_Y(v)^2]. \tag{123}$$

Thus the missing B_2 envelope is measurable as a supremum of fourth score moments along the declared chart. Proposition 6.7 separately supplies the third moment needed by nonzero-α connections and the exact cancellation $\Gamma^{(1)} = 0$ in natural affine-head coordinates.

The Bernoulli family in Proposition 6.9 proves that the extra condition cannot be discarded in general. More invariantly, because $g = d\theta^2$ there,

$$\|\nabla^{(\alpha)} - \nabla^{\mathrm{LC}}\|_g = |\alpha \cot\theta| \longrightarrow \infty. \tag{124}$$

Flat exponential or mixture coordinates do not contradict this statement: their coordinate changes become nonuniform near the boundary. The main theorems concern interior maps with no *uniform* probability floor. If exact zeros or changing support are admitted, smoothness of ψ must be imposed directly and the induced object is a Hellinger extension rather than an ordinary regular Fisher model on the open simplex.

A.2 Metric-compatible transport bound

Let $A = \nabla^{p,\mathrm{LC}} - \nabla^{\widetilde{p},\mathrm{LC}}$. Up to the sign convention for A, the exact variation-of-connection identity is

$$P^{p,\mathrm{LC}}_{1\leftarrow 0} - P^{\widetilde{p},\mathrm{LC}}_{1\leftarrow 0} = -\int_0^1 P^{p,\mathrm{LC}}_{1\leftarrow t}\, A_{\gamma(t)}\Big(\dot\gamma(t), P^{\widetilde{p},\mathrm{LC}}_{t\leftarrow 0}\,\cdot\Big)\,\mathrm{d}t. \tag{125}$$

Metric compatibility and $\lambda I \preceq g \preceq B_1^2 I$ imply

$$\|P^{\mathrm{LC}}_{t\leftarrow s}\|_{\mathrm{op}} \le \frac{B_1}{\sqrt{\lambda}} \tag{126}$$

for either geometry. Substitution into (125) gives the polynomial branch of (56).

For finite-scale reporting define

$$\tau_\gamma = \frac{B_1 B_2}{\lambda} L_\gamma, \qquad \kappa_{\mathrm{env}} = \frac{B_1^2}{\lambda}. \tag{127}$$

The transport theorem reads

$$\|P^{p,\mathrm{LC}}_\gamma - P^{\widetilde{p},\mathrm{LC}}_\gamma\|_{\mathrm{op}} \le L_\gamma D_{\mathrm{LC}} \min\{e^{\tau_\gamma}, \kappa_{\mathrm{env}}\}. \tag{128}$$

At one point, $\|D\psi\|_{\mathrm{op}}^2 = \lambda_{\max}(g)$, so pointwise-tight constants give $B_1^2/\lambda = \kappa(g)$. Uniform theorem constants instead give

$$\kappa_{\mathrm{env}} = \frac{\sup_K \lambda_{\max}(g)}{\inf_K \lambda_{\min}(g)} \geq \sup_K \kappa(g), \tag{129}$$

and equality need not hold. Both quantities are relative to the declared background chart. The existing pilot reports pointwise base-state spectra but not B_2, δ_1, δ_2, or pathwise extrema. It therefore motivates a conditioning audit but cannot imply a threshold such as $B_2 L_\gamma \lesssim 10^{-4}$ or establish that the theorem is intrinsically short-path. A numerical certificate requires measuring every term in (128) along the actual comparison path.

Code and manuscript availability

The manuscript source, model-free predictive-geometry implementation, CPU tests, and experimental protocols are available at

`https://github.com/guybass/probalistic-latent-spaces`.

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